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Procedia Computer Science 233 (2024) 733–742

Procedia Computer Science

www.elsevier.com/locate/procedia

5th International Conference on Innovative Data Communication Technologies and Application (ICIDCA 2024)

Real-Time Speaker Identification and Subtitle Overlay with Multithreaded Audio Video Processing

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Abstract

Active Speaker Detection is pivotal in a multitude of applications, particularly in processing live Audio-Video (AV) streams. Current implementations predominantly focus on processing saved video files, limiting their real-time applicability. Addressing this gap, the proposed model leverages a multi-threading-based system to detect active speakers in live AV streams. This system forms a critical component in an innovative software solution designed to generate real-time subtitles and elegantly overlay them aside from the active speaker. This feature is especially beneficial for individuals with hearing impairments and facilitates the transcription of foreign languages into English, thereby improving human interaction and understanding. Our approach stands out for its ability to process live AV streams promptly for immediate speaker identification and subtitle overlay, marking a significant advancement in real-time communication assistance.

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Peer-review under responsibility of the scientific committee of the 5th International Conference on Innovative Data Communication Technologies and Application

Keywords: TalkNet;AVSR; Multithreading;Subtitles;S3FD;Google STT API;Active Speaker

1. Introduction

In the contemporary global landscape, individuals with diverse linguistic backgrounds interact with video content and discourse delivered in diverse languages. This poses a challenge for people with hearing impairment or limited proficiency in spoken language. To ensure accessibility for a larger audience, the proposed project focuses on developing software designed to generate subtitles in real-time. This innovative tool not only adds subtitles to videos but also recognizes the speaker [19][9] and provides real-time translation, facilitating communication among individuals from various linguistic backgrounds.

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10.1016/j.procs.2024.03.262

Consider a scenario where an individual proficient in English encounters a video in a foreign language, the proposed method automatically generates English subtitles in real time, making the content more accessible. Moreover, this application proves invaluable for individuals with hearing impairments, enabling them to follow conversations through on-screen real-time generated subtitles. Consequently, the software significantly improves communication between individuals who do not share a common language, promoting efficiency and understanding in cross-linguistic relationships.

This research paper meticulously details the internal functionality of the software, delves into the technical components, evaluates its accuracy, and outlines the challenges encountered during its development. Furthermore, it explores potential avenues for improvement, underscoring the broader project goal of not only enhancing video accessibility but also advancing communication and global understanding in the digital age.

2. Related Works

2.1 Region of Interest for AVSR

The Hybrid Lip-Reading Network (HLR-Net) model [1] which has remarkable performance in the domain of lip-reading comprises three stages: preprocessing, encoder, and decoder stages, which utilize inception, gradient preservation, and bidirectional GRU layers to build the encoder and attention, and fully connected layer combined with activation functions to build the decoder stage, which performs Connectionist Temporal Classification (CTC). Utilizing a CNN model, the system processes a video capturing lip movements. It employs the OpenCV library to convert the video into individual frames and subsequently utilizes the Dlib library to extract the mouth region from these frames. The obtained frames undergo normalization to produce the final frame. This final frame is then inputted into the deep learning model, which generates subtitles as its output. Comparative analysis against three contemporary models, LipNet[2], e lip-reading model with cascaded attention (LCANet)[3], and attention-CTC (A-CTC)[4][17] on the GRID corpus dataset, reveals that HLR-Net excels for unseen speakers and for overlapped speakers, underscoring its potential in advancing the field of lip-reading

2.2 Active Speaker Detection

The research introduced TalkNet[5], an advanced system capable of making decisions by simultaneously considering short-term and long-term features[21].TalkNet is a system that processes videos with faces and their associated audio to determine whether a person, in each frame, is speaking. It is built with two components: a frontend with feature representation and a backend with speaker detection classification. This frontend is equipped with temporal encoders for both audio and video. These encoders transform audio and video signals, based on individual frames, into sequential embeddings that capture the temporal context. In the backend, a cross-attention mechanism facilitates dynamic alignment between audio and visual elements. Additionally, a self-attention mechanism is employed to analyze speaking activities within the temporal context on an utterance basis. Experimental findings reveal that TalkNet surpasses the performance of current state-of-the-art systems on both the AVA-ActiveSpeaker dataset and the Columbia ASD dataset, achieving significant enhancements of 3.5% and 2.2%, respectively.

2.3 Multithreading

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Multithreading is a key programming methodology that involves dividing a single process into smaller threads, each equipped to run independently within an application while sharing the same memory and resources. This approach greatly improves the efficiency and responsiveness of a program. In software development, multithreading proves highly beneficial by enabling optimal resource utilization when various components of a program operate

concurrently. This is particularly advantageous for independent tasks such as I/O operations, network processes, and computational tasks. The implementation of multithreading leads to a substantial enhancement in performance, allowing for effective resource utilization and ultimately contributing to superior software quality.

2.4 Subtitle Placement

In the study conducted by YONGTAO HU[6], researchers concentrated on investigating the effective placement of subtitles to enable viewers to easily follow both the video content and the subtitles simultaneously. The study introduces a framework that takes a video, along with its subtitle file in SRT format as input, and generates a version of the video where subtitles are positioned alongside their respective speakers. Speaker detection involves utilizing a lip motion technique, while face tracking is done using a frontal and profile Viola-Jones face detector [16]. Through experiments, it was found that this speaker detection method outperformed [13] by 8.5%–20.5% and [14] by 7.5%–17%.

. In [11][20], the study explores whether the inclusion of visual features alongside audio features can enhance the accuracy of machine learning models in speech-to-text processing. Recognition of Isolated Words involves utilizing mouth zernike moments on mouth images for feature extraction. The study reported 100% accuracy in audio recognition and 63.8% in video recognition.

2.5 Relevance of Real-Time Application

This model utilizes Google Speech-to-Text API to generate subtitles in real time, handling live audio-video streams. In contrast to previous systems that concentrated on pre-recorded videos with pre-prepared subtitles, this framework operates in real-time mode by overlaying subtitles on the speakers during the ongoing event sessions. The system accomplishes this by employing multithreading architectures, enabling the concurrent execution of multiple threads. This approach aids in efficiently processing the audio-video stream, automatically transcribing it into text, and simultaneously generating the corresponding subtitles. The utilization of multi-threading allows for the subdivision of tasks into smaller, manageable units, enhancing the system's responsiveness and efficiency in producing real-time subtitles. The framework seamlessly integrates live stream processing, online transcription, and multi-threading to provide viewers with instant captions while attending a live presentation, meeting, or event stream.

2.6 Novelty of proposed model established over existing models

Unlike conventional methods that rely on saved subtitle files, the proposed method seamlessly integrates live audio speech-to-text (STT) capabilities. Unlike current models, our system stands out in real-time speaker identification, dynamically adapting to speaker transitions to enhance subtitle accuracy. This real-time capability not only improves the model's adaptability with various content, such as live events but also eliminates the necessity for pre-existing subtitles, thereby enhancing accuracy and timeliness. By reducing dependences on prior datasets, our model distinguishes itself for its versatility and scalability, offering a cost-efficient solution.

3. Methodology

3.1 Problem Foundation

When working with saved files, processing time poses no challenges, but real-time subtitle generation and overlay require the implementation of a multi-threading system. This system concurrently processes new frames and generates transcribed text under active speakers. Unlike previous software design that exclusively dealt with saved files and

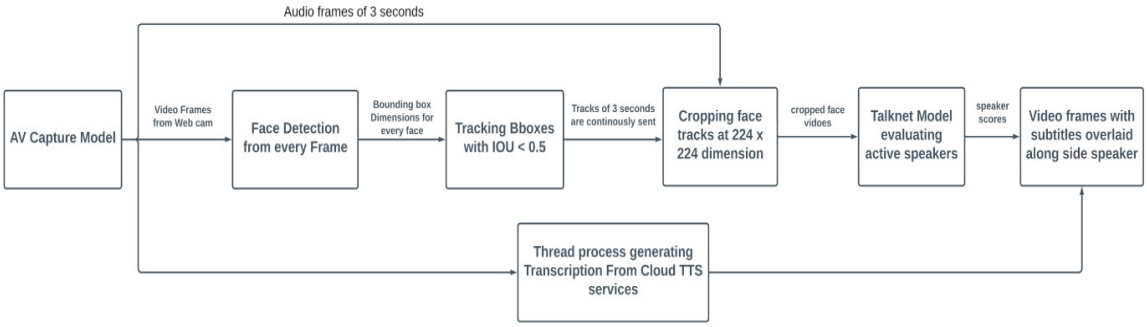


Fig. 1. Work Flow

had readily available subtitle files, the proposed approach necessitates integrating Speech-To-Text services to obtain real transcriptions before overlaying them on the respective speakers. The proposed method consists of the following components as shown in Fig.1

- Preprocessing of Generated Audio Video Frames
- Real-Time Detection of Active Speakers
- Real-Time Transcription and Translation
- Overlaying subtitles in an optimal position

3.2 Assumptions

The following assumptions underpin the functioning of the real-time subtitle generation system.

Assumption 1: In the context of real-time subtitle generation, the initial assumption is that speakers never speak over each other. This implies a requirement for clear and distinct speech from each speaker, avoiding overlapping or simultaneous talking .

Assumption 2: The second assumption relates to the language used by the speakers in the video, assuming that all speakers communicate in English.

Assumption 3: The final assumption is associated with visual aspects of the video content, assuming that the faces of the speakers are visible in the video stream.

3.3 Preliminaries

The model proposes to use multiple existing modules for various components of the workflow.

TalkNet is as a comprehensive end-to-end system. It processes a cropped facial video and the associated audio, determining whether an individual is speaking in each frame of the video. The system is structured with two main parts: a front-end for representing features, and a back-end classifier dedicated to speaker detection, as detailed in [5].

The model utilizes cropped videos of 224x224 resolution for facial detection within a video, adhering to a frame rate of 25 video frames and 100 audio frames per second, evaluating these frames and outputting scores for every face in every frame in the AV stream.

Concurrently during the execution of detecting active speakers [18], the model uses the audio frames in a background thread. The system utilizes state-of-the-art Google Cloud's Speech-To-Text v1 API to generate real-time English transcriptions. The transcription data is then incorporated into the pipeline to overlay speech transcriptions on the video frames once the Speaker Scores are calculated. In achieving this, the system must select an optimal position in the video next to the speaker.

To implement this in a real-time system, the system processes only consecutive time intervals of audio and video and uses multi-threading to compute different modules in parallel. As shown in Fig .1, the pipeline ensures the tasks are split and processed concurrently, significantly reducing the delay by eliminating the need for the entire 3-second AV stream to arrive completely. The processing of the AV streams after the initial three seconds involves appending our previous face tracks before sending them to the network for better evaluation.

3.4 Capturing Audio-Video Stream

This marks the initial stage in the workflow, the model employs the PyAudio library to capture audio frames at 16 kHz and the OpenCV module to capture 25 fps. These frames are then forwarded into threads running the other modules. The above configurations are based on their suitability for the TalkNet model to score active speakers.

3.5 Speaker Detection

The model performs speaker detection for every 3-second interval of the audio-video stream, aiming to identify active speakers in each video frame. As illustrated in Fig.1, the system initially detects faces and tracks them, generating cropped face video saved as AVI files. These videos are then inputted into TalkNet to obtain scores for every face in each frame of the provided video stream. Each step of this process is explained in detail in Algorithm 1.

Algorithm 1: Live Active Speaker Detection

Input: *Live Audio-Video Stream from Microphone and Camera.*

Output: *Detected Active Speaker(s)*

```
while stream is on do
    frame ← stream.read()
    Faces ← S3FDModel.track(frame)
    Scores ← []
    for face in Faces do
        score ← TalkNet.evaluate(audio, video)
        Scores.append(score)
    end
end
return Scores
```

3.5.1 Face Detection and Tracking

The system uses the PyTorch implementation of the S3FD [15] model for the face detection tasks. Utilizing the CUDA implementation, the system runs this model using NVIDIA GPUs to achieve faster inference. It then send the bounding boxes of the faces detected in each frame to the tracking thread.

The tracking thread monitors faces between adjacent frames by calculating the Intersection over Union of the area between the closest bounding boxes in consecutive frames. Bounding with an IoU greater than 0.5 are added as new bounding boxes and all tracks are sent to the cropping thread at the end of the 3-second time interval.

3.5.2 Generating Face Crops

The Bounding box tracks provided by the tracking thread are used to crop the original audio-video stream into face videos reshaped to a size of 224x224 pixels. These cropped videos are sent to the scoring thread, which runs inference on them using the TalkNet Model.

3.5.3 Evaluating the scores using TalkNet

The scoring thread feeds the output of the cropping thread into the TalkNet model which then returns frame-wise and face-wise scores for every speaker normalized at a range of -3.0 to 3.0. The scores are sent to the final visualization thread.

3.6 Transcription

The transcription thread utilizes the Google Speech-To-Text v1 API for real-time speech-to-text transcription. Online Text-to-Speech services enable us to transcribe text from multiple languages, and handle translation. For the scope of this project, the model exclusively utilizes English Speech-to-Text service. The transcriptions, along with the beginning and ending times corresponding to the audio source are forwarded to the visualization thread facilitating a time-bound overlay of the subtitles.

3.7 Subtitle Placement

The final step is handled by the visualization thread, which takes input from both the scoring thread and the transcription thread to identify active speakers and the subtitles to overlay, as outlined in Algorithm 2.

The optimal placement for subtitles involves positioning them in proximity to the corresponding speakers. Achieving this objective involves several considerations: (1) In each frame, subtitles should be situated close to their respective speakers to avoid any potential confusion regarding the speaker's identity; (2) Over successive frames, the collective distance between all subtitle placements should be minimized to reduce strain on the viewer's eyes; (3) Subtitles should avoid placement near the edges of the screen to prevent distraction from the central viewing area; (4) Subtitles should be positioned in a manner that does not obstruct essential visual elements, such as faces[6].

The model uses the OpenCV module to insert the subtitles in each frame near the active speaker. The frame number is used to determine the time at which the frame is being displayed, and the subtitles contain timestamps that assist in determining when to display each subtitle. The active speaker score helps decide under which speaker a given subtitle is to be displayed.

Algorithm 2: Overlay Subtitles

Input: *Subtitles with Timestamps and Faces with TalkNet scores.*

Output: *Subtitles Overlayed on Video*

```
subs ← subtitles with timestamps returned from STT module
scores ← Scores of each face in every video frame
overlay ← CV2 overlay module
for frame in video do
    time = frame[id]/25
    for subtitle, time in subs do
        if subtitle[beginTime] <= time <= subtitle[endTime] then
            overlay.add(face,subtitle)
    end
end
```

3.8 Further Considerations

In TV shows or movies, significant shot changes can lead to a speaker either appearing or disappearing. In such instances, the positioning of subtitles should adapt accordingly. For example, if a speaker disappears following a shot change, subtitles should be relocated to a default position, such as the bottom of the screen [6]. However, the system encounters this issue as the entire AV stream is recorded by the same camera. Any movement in the camera does not pose a problem, as the video time intervals reset the face track list and TalkNet evaluates new crops.

4. Experimental Results

4.1 System Specification

- Windows 11 i7 - 11th Gen Intel(R) Core (TM) i7-11800H @ 2.30GHz, RTX 3060, 6GB laptop GPU
- 1280 * 720 video
- 16000 kHz audio
- 25 FPS

As illustrated in Fig. 2(a), this corresponds to the latency levels of face detection at different intervals in milliseconds for different individuals appearing in the audio-visual stream. The latency for face detection exhibits an obvious upward trend as more people appear in the stream.

Fig.2(b) illustrates the performance time of the TalkNet model based on the number of people in an audio-visual stream. It portrays an individualistic trend in which the duration of processing is highly influenced by alterations in several users. The time spent processing any amount of people in a stream can vary because this suggests demand or resource needs as more people enter the stream. An empirical assessment of TalkNet's processing time towards crowds of varying sizes brings significant inputs to improving computing resources, hence leading to an effective response of the system in real-time identification and transcription of speakers

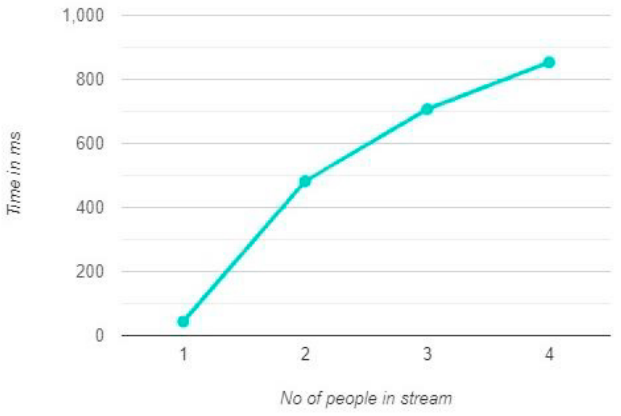


Fig. 2 . (a) Processing Latency

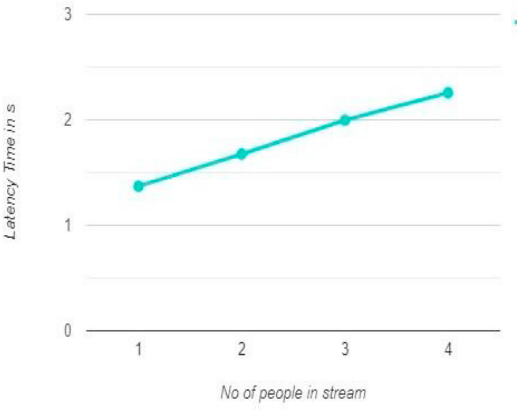


Fig.2. (b) TalkNet Performance

5. Output Images from the proposed software

The following are the outputs under different scenarios of the project as shown in Fig 3:

- When a visible speaker is speaking in the video, a green box will be drawn around the speaker with subtitles placed under them, and the non-speaker will be surrounded by a red box.
- When the speaker is outside the video frame, the subtitles will be placed at the bottom.

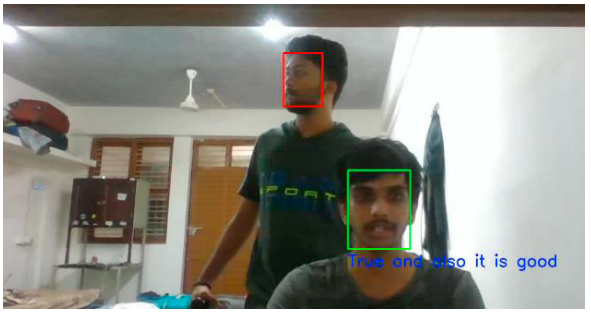


Fig.3. (a). Visible Speaker

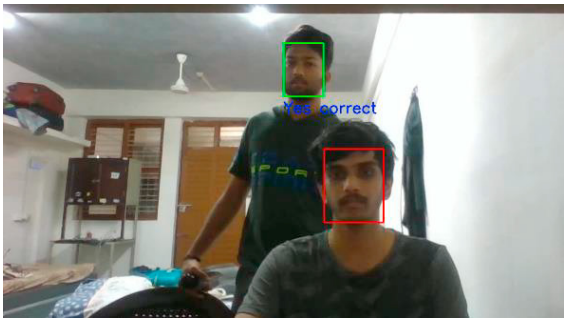


Fig .3.(b).Visible Speaker

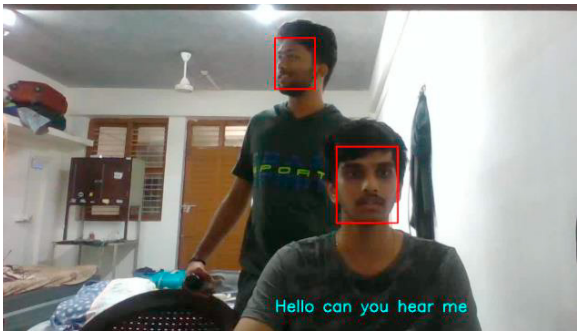


Fig 3. (b). Speaker outside Frame

Fig.3. Output Images

6. Discussion and Conclusion

The research presented a novel system for real-time speaker identification and subtitle overlay, utilizing a multithreading approach. The system's ability to process live audio-video streams distinguishes it from traditional methods dependent on saved subtitle files, offering real-time adaptability to various content types, including live events. The system exhibits proficiency in real-time processing and adjusts to speaker transitions, thereby enhancing the accuracy of subtitles. The research uses multithreading to improve computational efficiency and responsiveness.

The research contributes to real-time audio-video processing and accessibility technologies. The developed system improves communication for people with hearing impairments and also bridges linguistic gaps in real-time interactions. The integration of live speech-to-text capabilities and the dynamic identification of speakers advances the application of subtitle technology. The system can be applied in real-world settings such as online meetings, educational material, and live broadcasts. It also benefits individuals with hearing impairments advancing accessibility and communication technologies.

Future research will focus on optimizing the software for low-powered devices, enhancing language-transcribing capabilities, and broadening the range of supported languages. Efforts will also be directed towards improving subtitle visibility and readability in various visual environments, ensuring that the subtitles are clear, easy to follow, and non-intrusive. Overall, the planned improvements are geared towards ensuring the software's compatibility with lighter devices, bolstering its multilingual speech recognition and transcription abilities, adding translation features, and refining subtitle presentation for a more accessible and versatile user experience.

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